MOHAMED MAGDY ELATTAR

AI Engineer | Software Engineer | Machine Learning & Data Science

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PROFILE

Computer Science graduate and independent builder specializing in applied machine learning, agentic AI, and full-stack SaaS product development. Consistently the technical lead and sole architect on every project undertaken — from a production multi-tenant pharmacy SaaS platform to multi-agent AI systems, IoT platforms, and computer-vision models — with a track record of over 15 completed end-to-end builds. Strong foundation in Python, SQL, TensorFlow, LLM API integration, and cloud/IoT deployment, paired with a self-driven approach to shipping working products rather than isolated exercises.

INDEPENDENT PROJECTS & FREELANCE WORK

Founder / Lead Developer — Self-Directed *2023 – Present*

Meridian — Multi-Branch Pharmacy SaaS Platform *React, Supabase (Postgres), Vercel, Row-Level Security*

- Architected and built a production-grade, multi-tenant SaaS platform for Egyptian multi-branch pharmacy chains, migrating the system from an initial Firebase/Firestore prototype to a relational Postgres/Supabase stack to support complex role-based access and reporting.
- Designed a seven-tier role hierarchy enforced end-to-end with Postgres Row-Level Security (RLS), covering branch-, chain-, and account-level data isolation.
- Built a drug interaction checker (DDInter 2.0 + RxNorm) and a cross-sell/upsell engine on top of a 24,800+ product Egyptian drug database, plus GPS-based delivery tracking (OpenRouteService/Leaflet) and offline-resilient checkout.
- Ran a full security and performance audit (RLS gap analysis, unfiltered realtime subscriptions, cron/backup scheduling, capacity modeling) and produced a hardening roadmap raising the platform's readiness score from ~6/10 to a projected 8+/10.
- Developed the tiered pricing model and financial plan (per-branch pricing, break-even and margin analysis, VAT-threshold planning) and a competitive analysis against 7 Egyptian pharmacy-software incumbents to prepare the product for pilot sales.

LegalMind AI — Multi-Agent Legal Operating System *Python, Vector DB, LLM Orchestration*

- Designed a multi-agent architecture with 11 specialist agents and hybrid (vector + keyword) search for an AI legal assistant targeting MENA, US, UK, and EU law firms.
- Produced a full build specification, design-principles document, and a UAE-market test pack; evaluated GLM-5.2 for generative agent tasks and selected an embedding strategy (BGE) and OCR approach for document-heavy workflows.

ATLAS — Multi-Agent Autonomous AI System *MCP, Tiered Memory, DAG Orchestration*

- Designed a multi-agent orchestration system using MCP tool protocols, DAG-based Plan-Execute-Replan workflows, and tiered memory (short/long-term) for autonomous task execution and PC control.
- Built an interactive React dashboard prototype featuring a live memory map, a self-reorganizing task board, and a bilingual (Arabic/English) voice command console.

LifeOS — AI Life-Coaching Platform *FastAPI, Postgres + pgvector, Celery, Twilio, Claude API*

- Built a WhatsApp-native personal management system (tasks, journaling, fitness) backed by a vector-search memory store and an LLM-driven coaching layer.
- Shipped Phase 0–2 production code: database schema, FastAPI backend with a Twilio webhook, and a Celery beat scheduler for proactive check-ins.

FitAI — Adaptive Fitness & Nutrition App *React, TypeScript, Node/Express, Firebase*

- Built a mobile-first web app that generates personalized workout and meal plans using an LLM, combining survey data, uploaded body-composition PDFs, and live weather data.
- Shipped a v2 adaptive quarterly planning system with calendar-based tracking, monthly AI-driven plan adjustments, and a safety interrupt for injuries or goal changes.

IoT Fleet Monitoring Platform *Full-stack web app · role-based access · GPS mapping*

- Designed and led development of a multi-tenant monitoring platform for IoT field sensors (smart drains, smart bins) with live GPS mapping of deployed units.
- Architected role-based dashboards for 4 user types (Admin, Manager, Worker, Viewer) and automated stakeholder reporting; currently load-testing ahead of onboarding paying customers.

AI-Powered Google Workspace Assistant *n8n · Gmail & Forms automation*

- Built an automated assistant classifying incoming emails by intent (marketing, support, spam) and auto-generating replies, plus a Forms-to-stakeholder notification pipeline — running end-to-end without manual intervention.

FEATURED DATA SCIENCE & MACHINE LEARNING PROJECTS

- **Mask Detection (Computer Vision):** Trained a TensorFlow CNN to detect face masks in images, ~91% accuracy.
- **Earthquake Damage Prediction (Nepal):** Logistic regression and decision-tree models on structural data from SQLite, identifying and correcting discriminatory bias in the dataset.
- **Volatility Forecasting (India):** GARCH model over API-sourced stock data, served through a custom API.
- **Bankruptcy Prediction (Poland):** Random forest and gradient boosting models with resampling to address severe class imbalance.
- **Customer Segmentation (US):** K-means clustering with PCA, delivered via an interactive Plotly Dash dashboard.
- **Air Quality Forecasting (Nairobi) · Real Estate Regression (Buenos Aires) · Housing Price Analysis (Mexico, 21,000+ records):** ARMA time-series modeling, regression pipelines with missing-value imputation, and price-driver analysis.

GRADUATION PROJECT

Smart Drain — IoT Structural Health Monitoring *2024*

- Designed an IoT system measuring street-drain water levels for early flood/damage detection, with multi-level sensors, color-coded alerting, and MQTT-based real-time transmission to a Firebase-backed dashboard, powered by solar + battery backup.

EDUCATION

Bachelor of Computer Science *Arab Academy for Science, Technology & Maritime Transport | 2020 – 2024 | GPA: 3.1 (Very Good)*

Diploma in Machine Learning & Artificial Intelligence *Amit Learning | 2023 – 2024 | Grade: Excellent (95%)*

CERTIFICATIONS

- Machine Learning & AI — Amit Learning (2024)
- AI Lab: Deep Learning for Computer Vision — WorldQuant University (07/2024 – 12/2024); CNNs, image classification, object detection, and facial recognition with TensorFlow and PyTorch.

TECHNICAL SKILLS

- **Languages:** Python, Java, SQL, TypeScript/JavaScript
- **AI / LLM Engineering:** LLM API integration (Claude, GLM), multi-agent orchestration, prompt engineering, MCP tool protocols, vector search (pgvector), RAG
- **ML / DL:** TensorFlow, PyTorch, scikit-learn, CNNs, time-series (ARMA/GARCH), classical ML (regression, trees, ensembles, clustering)
- **Backend / Data:** FastAPI, Node/Express, PostgreSQL (Supabase, Row-Level Security), Firebase/Firestore, MongoDB, SQLite, Celery, pandas, ETL pipelines
- **Frontend:** React, Vite, JavaFX, Streamlit, Plotly Dash
- **Systems / Deployment:** Vercel, Render, MQTT/IoT, REST APIs, Twilio
- **Automation:** n8n workflow automation, Google Workspace API (Gmail, Forms)

LANGUAGES

- Arabic — Native
- English — Intermediate